# Lesson 4. Systems of Differential Equations I

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# 1 Basic Theory of Systems of ODEs

## 1.1 Motivating Examples

**Example 1.1** (Damped Harmonic Oscillator)**.** Consider the equation for a damped harmonic oscillator:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0,$$

where $m$ is the mass, $x(t)$ is the displacement, $b$ is the damping coefficient, and $k$ is the spring constant.

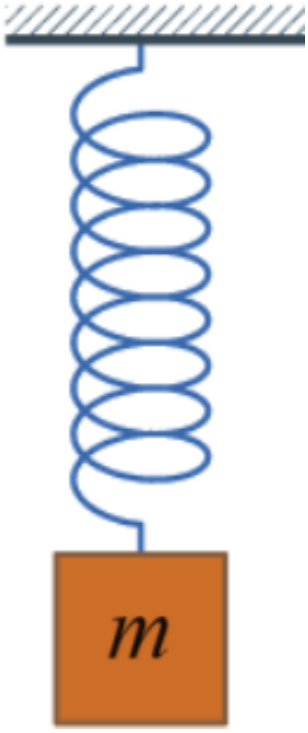


Figure 1: Damped harmonic oscillator.

This second-order ODE can be rewritten as a **system** of first-order ODEs using **substitution**. We introduce new variables:

$$x_1 = x \quad \text{and} \quad x_2 = \frac{dx}{dt}.$$

Then the original equation transforms into:

$$\begin{cases} \dfrac{dx_1}{dt} = x_2, \\ \dfrac{dx_2}{dt} = -\dfrac{b}{m}\,x_2 - \dfrac{k}{m}\,x_1. \end{cases}$$

Here, $x_1$ represents the position and $x_2$ represents the velocity. This system provides an equivalent description of the dynamics of the damped harmonic oscillator.

**Example 1.2** (Simple Pendulum)**.** The swinging of a pendulum is governed by the equation:

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0,$$

where $x(t)$ is the angular displacement, $g$ is the acceleration due to gravity, and $L$ is the length of the pendulum.

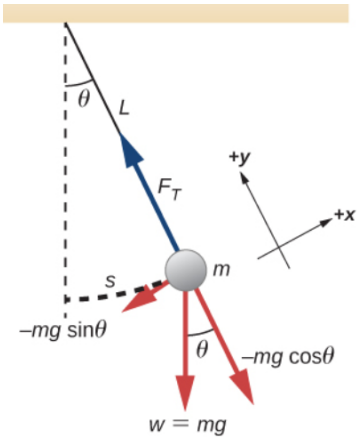


Figure 2: Simple pendulum.

Introducing new variables $x_1 = x$ and $x_2 = \frac{dx}{dt}$, we obtain the system:

$$\begin{cases} \dfrac{dx_1}{dt} = x_2, \\ \dfrac{dx_2}{dt} = -\dfrac{g}{L}\sin(x_1). \end{cases}$$

Here, $x_1$ represents the angular displacement and $x_2$ represents the angular velocity.

## 1.2 Systems of Differential Equations

**Definition 1.1** (System of Differential Equations)**.** A **system of differential equations** is a finite set of differential equations. Systems can be classified by:

- type: systems of ODEs or PDEs;
- order and degree;
- linearity.

**Definition 1.2** (Standard Form)**.** A system of first-order ordinary differential equations can be expressed in the **standard form**:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

A system of ODEs is said to be **linear** if each function $f_i$ is linear in all the dependent variables $x_1, x_2, \ldots, x_n$. A system is **nonlinear** if any of the functions $f_i$ involve nonlinear terms in the dependent variables.

**Example 1.3** (Linear System)**.** The damped harmonic oscillator system:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\dfrac{k}{m}\, x_1 - \dfrac{b}{m}\, x_2 \end{cases}$$

is linear, since the right-hand sides are linear functions of $x_1$ and $x_2$.

**Example 1.4** (Nonlinear System)**.** The simple pendulum system:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\dfrac{g}{L} \sin(x_1) \end{cases}$$

is nonlinear, due to the $\sin(x_1)$ term.

**Definition 1.3** (Solution)**.** A system of functions

$$\begin{cases} x_1(t) = \Phi_1(t) \\ x_2(t) = \Phi_2(t) \\ \vdots \\ x_n(t) = \Phi_n(t) \end{cases}$$

is a **solution** to the system of differential equations if each function $\Phi_j(t)$ for $1 \leq j \leq n$ is continuously differentiable on an interval $I$ and satisfies the corresponding differential equation on $I$.

## 1.3 Initial Value Problems for Systems

**Definition 1.4** (IVP for Systems)**.** An **initial value problem (IVP)** for a system of first-order ODEs consists of the system of differential equations together with initial conditions:

$$\begin{cases} x_1(t_0) = k_1 \\ x_2(t_0) = k_2 \\ \vdots \\ x_n(t_0) = k_n \end{cases}$$

where $t_0$ is the initial time and $k_1, k_2, \ldots, k_n$ are the initial values of the functions at $t = t_0$.

A **solution to the IVP** is a set of functions $\{x_1(t), x_2(t), \ldots, x_n(t)\}$ that satisfies both the system of differential equations and the initial conditions for $t$ in some interval containing $t_0$.

**Theorem 1.1** (Existence and Uniqueness)**.** *Let*

$$f_i, \quad 1 \leq i \leq n \quad \textit{and} \quad \frac{\partial f_i}{\partial x_j}, \quad 1 \leq i \leq n,\ 1 \leq j \leq n$$

*be continuous in some domain $R$ containing the point $(t_0, k_1, \ldots, k_n)$. Then the initial value problem has a unique solution on some interval $(t_0 - \alpha, t_0 + \alpha)$.*

**Theorem 1.2** (Existence and Uniqueness for Linear Systems)**.** *Consider the linear system*

$$\mathbf{x}'(t) = A(t)\mathbf{x}(t) + \mathbf{b}(t).$$

*If the functions $a_{ij}(t)$ and $b_i(t)$ are continuous on an interval $(t_0 - \alpha, t_0 + \alpha)$, then the initial value problem $\mathbf{x}(t_0) = \mathbf{x}_0$ has a unique solution on this interval.*

## 1.4 Phase Space and Qualitative Analysis

The solution space is called a **phase space** (or **phase portrait**), and the solution curve $\mathbf{x} = (x_1(t), \ldots, x_n(t))$ is called the **trajectory** or the **orbit** in the phase space $\mathbb{R}^n$.

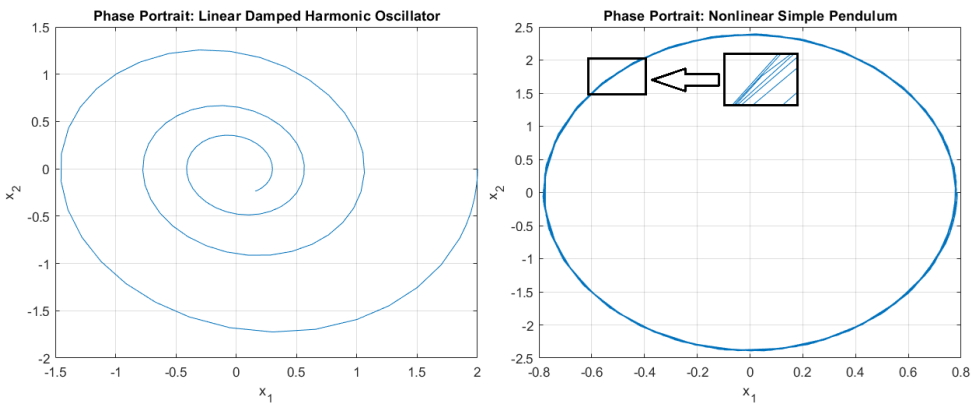


Figure 3: Phase space: solution curves in the $x_1x_2$-plane and individual component plots.

**Qualitative analysis** of differential equations is a powerful method for understanding the behavior of systems without needing to solve the equations explicitly. The main steps include:

1. Finding equilibrium points;
2. Determining stability;
3. Global behavior analysis;
4. Identifying the phase space and sketching trajectories.

# 2 Equilibrium Points and Stability Theory

## 2.1 Equilibrium Points

**Definition 2.1** (Equilibrium Point)**. Equilibrium points** (also called **critical points** or **stationary points**) of a system of differential equations are points in the phase space where the system does not change over time. For a system

$$\begin{cases} x_1' = f_1(x_1, x_2, \ldots, x_n) \\ x_2' = f_2(x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(x_1, x_2, \ldots, x_n) \end{cases}$$

an equilibrium point $\mathbf{x}^* = (x_1^*, x_2^*, \ldots, x_n^*)$ satisfies:

$$f_1(\mathbf{x}^*) = 0, \quad f_2(\mathbf{x}^*) = 0, \quad \ldots, \quad f_n(\mathbf{x}^*) = 0.$$

Equilibrium points represent the states where the system is in balance: once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

**Example 2.1** (Equilibrium of the Damped Harmonic Oscillator)**.** For the linear system

$$\begin{cases} x_1' = x_2 \\ x_2' = -\dfrac{k}{m}\, x_1 - \dfrac{b}{m}\, x_2 \end{cases}$$

setting $x_1' = 0$ and $x_2' = 0$ gives $x_2 = 0$ and $-\frac{k}{m}x_1 = 0$, so $x_1 = 0$. The unique equilibrium point is $(0, 0)$.

Physically, this represents the state where the mass is at rest in its natural position (neither stretched nor compressed) and not moving. All forces are balanced: the spring force and the damping force cancel each other out.

**Example 2.2** (Equilibria of the Simple Pendulum)**.** For the nonlinear system

$$\begin{cases} x_1' = x_2 \\ x_2' = -\dfrac{g}{L} \sin(x_1) \end{cases}$$

setting $x_1' = 0$ and $x_2' = 0$ gives $x_2 = 0$ and $\sin(x_1) = 0$, so $x_1 = \pi n$ for integer $n$. The equilibrium points are $(\pi n, 0)$.

- For $n$ even (e.g., $x_1 = 0, 2\pi$): the pendulum hangs straight down. This is a **stable equilibrium**—a small push causes it to swing back.
- For $n$ odd (e.g., $x_1 = \pi$): the pendulum points straight up. This is an **unstable equilibrium**—a small disturbance causes it to fall away.

These equilibria represent states where the pendulum has no kinetic energy (not moving) and potential energy is either minimized (stable, pendulum down) or maximized (unstable, pendulum up).

## 2.2 Lyapunov Stability

Consider the system of DEs in vector form:

$$\mathbf{X}' = \mathbf{f}(t, \mathbf{X}), \quad \mathbf{X}(0) = \mathbf{X}_0. \tag{1}$$

**Definition 2.2** (Lyapunov Stability)**.** The solution $\boldsymbol{\varphi}(t)$ of the IVP (1) is said to be **stable in the sense of Lyapunov** if for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that if

$$|\mathbf{X}(0) - \boldsymbol{\varphi}(0)| < \delta,$$

then

$$|\mathbf{X}(t) - \boldsymbol{\varphi}(t)| < \varepsilon$$

for all $t \geq 0$. Otherwise, the solution $\boldsymbol{\varphi}(t)$ is said to be **unstable**.

In the case $n = 2$, Lyapunov stability means that any trajectory $\mathbf{X}(t)$ starting in a $\delta(\varepsilon)$-neighborhood of $\boldsymbol{\varphi}(0)$ remains inside the tube with maximum radius $\varepsilon$ for all $t \geq 0$.

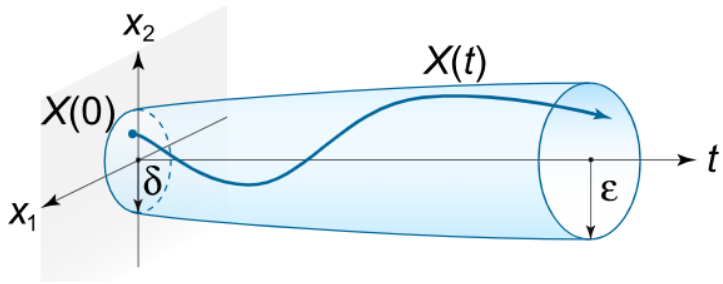


Figure 4: Lyapunov stability: trajectories remain within the $\varepsilon$-tube.

### 2.3 Asymptotic Stability

**Definition 2.3** (Asymptotic Stability)**.** If the solution $\boldsymbol{\varphi}(t)$ of the IVP (1) is not only stable in the sense of Lyapunov, but also satisfies

$$\lim_{t\to\infty} |\mathbf{X}(t) - \boldsymbol{\varphi}(t)| = 0,$$

provided that $|\mathbf{X}(0) - \boldsymbol{\varphi}(0)| < \delta$, then we say that the solution $\boldsymbol{\varphi}(t)$ is **asymptotically stable**.

In the case $n = 2$, asymptotic stability means that all solutions sufficiently close to $\boldsymbol{\varphi}(0)$ at the initial time gradually converge to $\boldsymbol{\varphi}(t)$ as $t \to \infty$.

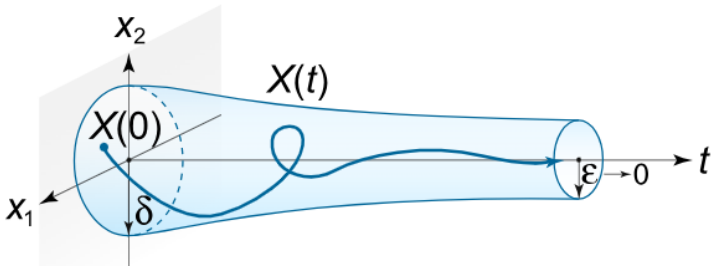


Figure 5: Asymptotic stability: trajectories converge to the solution.

## 3 Equilibrium Points of Linear Autonomous Systems

### 3.1 Matrix Form of Linear Systems

When the equations in the system are linear, matrix algebra provides a compact notation. Introduce the column vectors and matrix:

$$\mathbf{x} = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix}, \quad A = \begin{bmatrix} a_{11}(t) & \cdots & a_{1n}(t) \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \cdots & a_{nn}(t) \end{bmatrix}.$$

Then the linear system can be written in the compact **matrix form**:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}. \tag{2}$$

A system is in **normal form** if it is expressed as $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$. It is called **homogeneous** when $\mathbf{f} = \mathbf{0}$; otherwise, it is **nonhomogeneous**. When the elements of $A$ are all constants, the system is said to have **constant coefficients**.

## 3.2 Homogeneous Systems with Constant Coefficients

Consider a first-order linear homogeneous system with constant coefficients in 2D:

$$\begin{cases} \dfrac{dx}{dt} = a_{11}x + a_{12}y \\ \dfrac{dy}{dt} = a_{21}x + a_{22}y \end{cases}$$

This system is **autonomous** since the right-hand sides do not explicitly contain the independent variable $t$.

The equilibrium positions can be found by solving $A\mathbf{X} = \mathbf{0}$. This equation has:

- the unique solution $\mathbf{X} = \mathbf{0}$ if $\det A \neq 0$;
- infinitely many equilibrium points if $\det A = 0$.

The classification of the equilibrium point at the origin is determined by the eigenvalues $\lambda_1, \lambda_2$ of the matrix $A$. When $\det A \neq 0$, there are four types:

| | **Equilibrium Type** | **Eigenvalues** $\lambda_1, \lambda_2$ |
|---|---|---|
| 1 | Node | Real, same sign |
| 2 | Saddle | Real, nonzero, opposite sign |
| 3 | Focus | Complex conjugate; nonzero real parts |
| 4 | Center | Purely imaginary |

## 3.3 Stability Theorems

The linear system $\mathbf{X}' = A(t)\mathbf{X} + \mathbf{f}(t)$ is stable if all its solutions are stable in the sense of Lyapunov. The nonhomogeneous system is stable with any free term $\mathbf{f}(t)$ if the zero solution of the associated homogeneous system $\mathbf{X}' = A(t)\mathbf{X}$ is stable. Therefore, when investigating stability in the class of linear systems, it is sufficient to analyze the homogeneous system.

Consider the homogeneous linear system

$$\mathbf{X}' = A\mathbf{X},$$

where $A$ is a constant $n \times n$ matrix. This autonomous system has the zero solution $\mathbf{X}(t) = \mathbf{0}$.

**Theorem 3.1** (Lyapunov Stability)**.** *A linear homogeneous system with constant coefficients is stable in the sense of Lyapunov if and only if all eigenvalues $\lambda_i$ of $A$ satisfy*

$$\mathrm{Re}[\lambda_i] \leq 0 \quad (i = 1, 2, \ldots, n).$$

*If the real part of an eigenvalue is equal to zero, the algebraic and geometric multiplicity of the eigenvalue must be the same (i.e., the corresponding Jordan block must be of size $1 \times 1$).*

**Theorem 3.2** (Asymptotic Stability)**.** *A linear homogeneous system with constant coefficients is asymptotically stable if and only if all eigenvalues $\lambda_i$ have negative real parts:*

$$\mathrm{Re}[\lambda_i] < 0 \quad (i = 1, 2, \ldots, n).$$

**Theorem 3.3** (Instability)**.** *A linear homogeneous system with constant coefficients is unstable if at least one of the following conditions is satisfied:*

- *The matrix $A$ has an eigenvalue $\lambda_i$ with a positive real part;*
- *The matrix $A$ has an eigenvalue $\lambda_i$ with zero real part, and the geometric multiplicity of $\lambda_i$ is less than its algebraic multiplicity.*

## 3.4 How to Sketch Phase Portraits

To sketch a phase portrait:

1. Find the eigenvalues of the matrix $A$.
2. Determine the type of the equilibrium point and the character of stability.
3. Find the equations of the isoclines:

$$\frac{dx}{dt} = a_{11}x + a_{12}y = 0 \quad \text{(vertical isocline)},$$

$$\frac{dy}{dt} = a_{21}x + a_{22}y = 0 \quad \text{(horizontal isocline)}.$$

4. If the equilibrium is a node or a saddle, compute the eigenvectors and draw the asymptotes parallel to the eigenvectors passing through the origin.
5. Schematically draw the phase portrait.
6. Show the direction of motion along the phase trajectories. In the case of a focus or center, determine the direction of twisting.

## 3.5 Phase Portrait Examples

**Example 3.1** (Stable Node)**.** Investigate the equilibrium positions and draw the phase portrait of:

$$\begin{cases} \dfrac{dx_1}{dt} = -x_1 \\ \dfrac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

**Step 1.** The system matrix and its determinant:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix}, \quad \det A = (-1)(-2) - 0 = 2 \neq 0.$$

Since $\det A \neq 0$, the system has the unique equilibrium point $\mathbf{X} = \mathbf{0}$.

The eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} -1-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} = (\lambda+1)(\lambda+2) = 0 \implies \lambda_1 = -1,\ \lambda_2 = -2.$$

**Step 2.** Both eigenvalues are real and negative, so the equilibrium point is a **stable node**.

**Step 3.** Isoclines:

- Vertical isocline: $-x_1 = 0$, i.e., $x_1 = 0$ (the $x_2$-axis).
- Horizontal isocline: $2x_1 - 2x_2 = 0$, i.e., $x_2 = x_1$.

**Step 4.** Eigenvectors:

- For $\lambda_1 = -1$:

$$(A - \lambda_1 I)\mathbf{V}_1 = \begin{bmatrix} 0 & 0 \\ 2 & -1 \end{bmatrix} \mathbf{V}_1 = \mathbf{0} \implies 2V_{11} - V_{21} = 0 \implies \mathbf{V}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

- For $\lambda_2 = -2$:

$$(A - \lambda_2 I)\mathbf{V}_2 = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \mathbf{V}_2 = \mathbf{0} \implies V_{12} = 0 \implies \mathbf{V}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

**Step 5.** The phase trajectories approach the origin tangent to the line directed along $\mathbf{V}_1$, as this eigenvector corresponds to the smallest (in absolute value) eigenvalue: $|\lambda_1| = 1$.

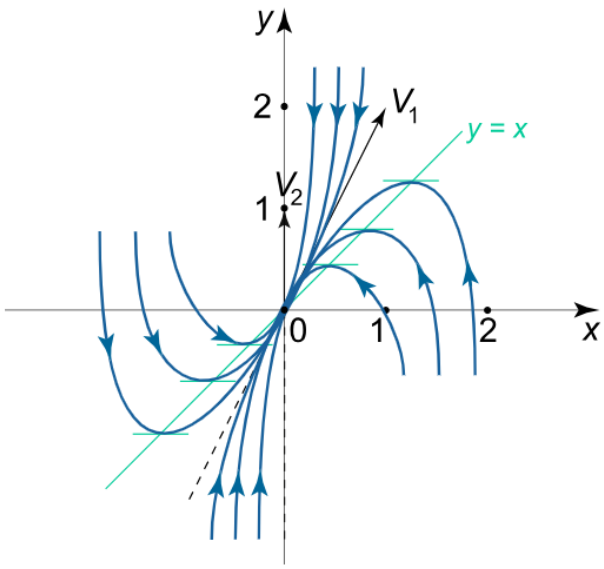


Figure 6: Phase portrait of the stable node.

**Example 3.2** (Saddle)**.** Investigate the equilibrium positions and draw the phase portrait of:

$$\begin{cases} \dfrac{dx_1}{dt} = x_1 + 3x_2 \\ \dfrac{dx_2}{dt} = 2x_1 \end{cases}$$

**Step 1.** The system matrix:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}, \quad \det A = -6 \neq 0.$$

The system has the unique equilibrium point $\mathbf{X} = \mathbf{0}$.

The eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 3 \\ 2 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 6 = 0 \implies \lambda_1 = 3,\ \lambda_2 = -2.$$

**Step 2.** The eigenvalues are real with opposite signs, so the equilibrium point is a **saddle** (unstable).

**Step 3.** Isoclines:

- Vertical isocline: $x_1 + 3x_2 = 0$, i.e., $x_2 = -\frac{x_1}{3}$.
- Horizontal isocline: $2x_1 = 0$, i.e., $x_1 = 0$ (the $x_2$-axis).

**Step 4.** The separatrices have the form $x_2 = kx_1$. Substituting into the system:

$$2x_1 = k(x_1 + 3kx_1) \implies 3k^2 + k - 2 = 0 \implies k_1 = -1,\ k_2 = \frac{2}{3}.$$

**Step 5.** To determine the direction of motion, evaluate $\frac{dx_2}{dt}$ at the point $(1, 0)$:

$$\frac{dx_2}{dt}(1, 0) = 2 \cdot 1 = 2 > 0.$$

The trajectory crosses the $x_1$-axis in the upward direction.

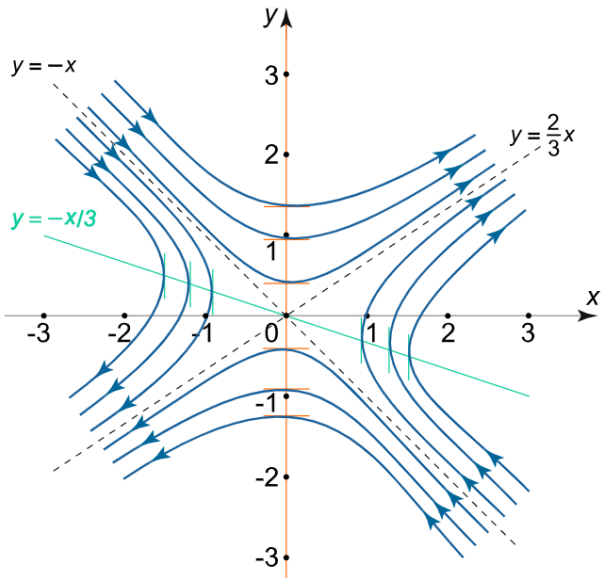


Figure 7: Phase portrait of the saddle.

**Example 3.3** (Unstable Focus)**.** Investigate the equilibrium positions and draw the phase portrait of:

$$\begin{cases} \dfrac{dx_1}{dt} = 3x_1 - 4x_2 \\ \dfrac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

**Step 1.** The system matrix:

$$A = \begin{bmatrix} 3 & -4 \\ 2 & -1 \end{bmatrix}, \quad \det A = -3 + 8 = 5 \neq 0.$$

The system has the unique equilibrium point $(0, 0)$.

**Step 2.** The eigenvalues:

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & -4 \\ 2 & -1-\lambda \end{vmatrix} = \lambda^2 - 2\lambda + 5 = 0.$$

The discriminant is $D = 4 - 20 = -16$, so

$$\lambda_{1,2} = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

The eigenvalues are complex conjugate with a positive real part, so the equilibrium is an **unstable focus**.

**Step 3.** Isoclines:

- Vertical isocline: $3x_1 - 4x_2 = 0$, i.e., $x_2 = \frac{3}{4}x_1$.
- Horizontal isocline: $2x_1 - x_2 = 0$, i.e., $x_2 = 2x_1$.

**Step 4.** To find the direction of twisting, evaluate $\frac{dx_2}{dt}$ at $(1, 0)$:

$$\frac{dx_2}{dt}(1, 0) = 2 \cdot 1 - 0 = 2 > 0.$$

Thus, the spirals twist counterclockwise.

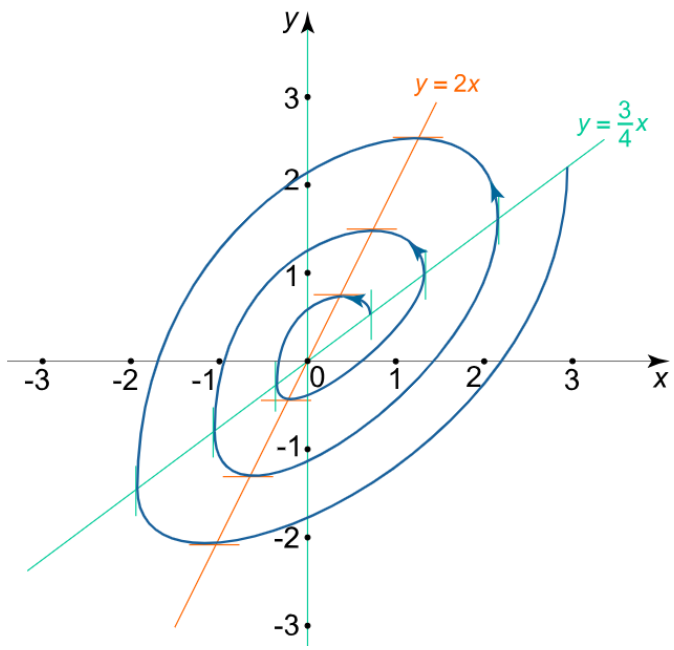


Figure 8: Phase portrait of the unstable focus.